HW2 Solutions

ECE 310: Digital Signal Processing, Fall 2026 Version: **1.0**

Due Date: 09/11/2026
Prepared by: Nick & Mia

1

Solution:

a) $y[n] = y[n-5] + x[n] + 10x[n-1]$

- **Linearity:** Let $T(x[n]) = x[n] + 10x[n-1]$ be the system. We test for superposition:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= (ax_1[n] + bx_2[n]) + 10(ax_1[n-1] + bx_2[n-1]) \\ &= a(x_1[n] + 10x_1[n-1]) + b(x_2[n] + 10x_2[n-1]) \\ &= a(y_1[n] - y_1[n-5]) + b(y_2[n] - y_2[n-5]) \end{aligned}$$

  The system is **linear**.

- **Time invariance:** Let $y'[n] - y'[n-5] = T(x[n-n_0]) = x[n-n_0] + 10x[n-n_0-1]$.
  The shifted output is $y[n-n_0] - y[n-n_0-5] = x[n-n_0] + 10x[(n-n_0)-1]$.
  Since $y'[n] - y'[n-5] = y[n-n_0] - y[n-n_0-5]$, the system is **time-invariant**.

b) $y[n-2] + 2y[n] = \cos(\frac{\pi}{6}n)x[n]$

- **Linearity:** Let $T(x[n]) = \cos(\frac{\pi}{6}n)x[n]$ be the system. We test for superposition:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= \cos(\frac{\pi}{6}n)(ax_1[n] + bx_2[n]) \\ &= a\cos(\frac{\pi}{6}n)x_1[n] + b\cos(\frac{\pi}{6}n)x_2[n] \\ &= a(y_1[n-2] + 2y_1[n]) + b(y_2[n-2] + 2y_2[n]) \end{aligned}$$

  The system is **linear**.

- **Time invariance:** Let $x[n] = \delta[n]$ and $x'[n] = \delta[n-3]$.
  Then $y[(n-2)-3] + 2y[(n)-3] = \delta[(n)-3]$ and $y'[n-2] + 2y'[n] = 0$.
  Since $y[(n-2)-3] + 2y[(n)-3] \neq y'[n-2] + 2y'[n]$, the system is **time-varying**.

c) $y[n] = \frac{1}{2}^{|n|}x[n]$

- **Linearity:** Let $T(x[n]) = \frac{1}{2}^{|n|}x[n]$ be the system. We test for superposition:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= \frac{1}{2}^{|n|}(ax_1[n] + bx_2[n]) \\ &= a\frac{1}{2}^{|n|}x_1[n] + b\frac{1}{2}^{|n|}x_2[n] \\ &= ay_1[n] + by_2[n] \end{aligned}$$

  The system is **linear**.

- **Time invariance:** Let $x[n] = \delta[n]$ and $x'[n] = \delta[n-1]$.
  Then $y[(n)-1] = \delta[(n)-1]$ and $y'[n] = \frac{1}{2}\delta[n-1]$.
  Since $y[(n)-1] \neq y'[n]$, the system is **time-varying**.

Grading: *18 points, 6 points each*

- For each of (a), (b), (c):
  - +3: correctly identified linearity with proper justification
  - +3: correctly identified time-variance/invariance with proper justification

2

Solution:

- **Linearity:** Let $T(x[n]) = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$ be the system. We test for superposition:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= \sum_{k=-\infty}^{\infty} (ax_1[k] + bx_2[k])h[n-k] \\ &= a\sum_{k=-\infty}^{\infty} x_1[k]h[n-k] + b\sum_{k=-\infty}^{\infty} x_2[k]h[n-k] \\ &= ay_1[n] + by_2[n] \end{aligned}$$

  The system is **linear**.

- **Shift invariance:** Let $T(x[n]) = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$ be the system and let $y'[n] = T(x[n-n_0]) = \sum_{k=-\infty}^{\infty} h[k]x[n-k-n_0] = \sum_{k=-\infty}^{\infty} h[k]x[(n-n_0)-k]$
  The shifted output is $y[n-n_0] = \sum_{k=-\infty}^{\infty} h[k]x[(n-n_0)-k]$.
  Since $y'[n] = y[n-n_0]$, the system is **shift-invariant**.

Grading: *20 points*

- +10: correctly proved linearity with proper justification
- +10: correctly proved time-variance/invariance with proper justification
- +0: empty or invalid

3

Solution:

First note that $\delta[n] = u[n] - u[n-1]$.

$$\begin{aligned} h[n] &= h[n] * \delta[n] \\ &= h[n] * (u[n] - u[n-1]) \\ &= h[n] * u[n] - h[n] * u[n-1] \end{aligned}$$

Then $y[n] = h[n] * x[n]$

$$\begin{aligned} y[n] &= h[n] * x[n] \\ &= (h[n] * u[n] - h[n] * u[n-1]) * x[n] \\ &= (h[n] * u[n]) * x[n] - (h[n] * u[n-1]) * x[n] \\ &= (h[n] * u[n]) * x[n] - h[n] * (u[n-1] * x[n]) \\ &= (h[n] * u[n]) * x[n] - h[n] * (u[n] * x[n-1]) \\ &= (h[n] * u[n]) * x[n] - (h[n] * u[n]) * x[n-1] \\ &= g[n] * x[n] - g[n] * x[n-1] \\ &= g[n] * (x[n] - x[n-1]) \end{aligned}$$

$$\boxed{y[n] = g[n] * (x[n] - x[n-1])}$$

Grading: *20 points*

- +20: correct answer
- +10: minor mistake with clear reasoning/thought process
- +0: empty or invalid

4

Solution:

First note that

$$\begin{aligned} x[n] - 3^{(-1)}x[n-1] &= 3^{(-n)}u[n] - 3^{(-1)}3^{(-(n-1))}u[n-1] \\ &= 3^{(-n)}u[n] - 3^{(-n)}u[n-1] \\ &= 3^{(-n)}(u[n] - u[n-1]) \\ &= 3^{(-n)}\delta[n] \\ &= \delta[n] \end{aligned}$$

Therefore by linearity and shift invariance

$$\begin{aligned} y[n] - 3^{(-1)}y[n-1] &= h[n] * (x[n] - 3^{(-1)}x[n-1]) \\ &= h[n] * \delta[n] \\ &= h[n] \end{aligned}$$

Then

$$\begin{aligned} y[n] - 3^{(-1)}y[n-1] &= \frac{1}{5^n}u[n-1] - \frac{3^{(-1)}}{5^{n-1}}u[n-2] \\ &= \frac{1}{5^n}\left(u[n-1] - \frac{5}{3}u[n-2]\right) \end{aligned}$$

$$\boxed{h[n] = \frac{1}{5^n}\left(u[n-1] - \frac{5}{3}u[n-2]\right)}$$

Grading: *22 points*

- +22: correct answer
- +16: minor math mistake with correct LTI reasoning
- +8: correctly related $\delta[n]$ and $u[n]$ but failed to apply superposition properly
- +0: empty or invalid

5

Solution:

a) $x[n] = \{-1, \underset{\uparrow}{0}, 1\}, h[n] = \{\underset{\uparrow}{1}, 2, 3, 4, 5\}$

We use matrix-vector multiplication where the matrix is formed by shifts of $h[n]$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \\ 4 & 3 & 2 \\ 5 & 4 & 3 \\ 0 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ -2 \\ -2 \\ -2 \\ 4 \\ 5 \end{bmatrix}$$

$$\boxed{x[n] * h[n] = \{-1, \underset{\uparrow}{-2}, -2, -2, -2, 4, 5\}}$$

b) $x[n] = 3^{(-n)}u[n], h[n] = \{\underset{\uparrow}{0}, 1, 2\}$

First, $h[n] = \delta[n-1] + 2\delta[n-2]$

Therefore by linearity and shift-invariance, $x[n] * h[n] = 3^{(-(n-1))}u[n-1] + 2 \cdot 3^{(-(n-2))}u[n-2]$. Which simplifies to $x[n] * h[n] = 3^{(-(n-1))}(u[n-1] + 6u[n-2])$

$$\boxed{x[n] * h[n] = 3^{(-(n-1))}(u[n-1] + 6u[n-2])}$$

c) $x[n] = u[n], h[n] = n(u[n] - u[n-4])$

Rewriting $h[n]$ as a sum of impulses:

$$h[n] = \delta[n-1] + 2\delta[n-2] + 3\delta[n-3]$$

Therefore by linearity and shift-invariance, $x[n] * h[n] = u[n-1] + 2u[n-2] + 3u[n-3]$

$$\boxed{x[n] * h[n] = u[n-1] + 2u[n-2] + 3u[n-3]}$$

d) $x[n] = (-1)^{(-n)}u[n], h[n] = e^{(-n)}u[n]$

Note that for integer k, $(-1)^{(-k)} = (-1)^k$

$$\begin{aligned} x[n] * h[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= \sum_{k=-\infty}^{\infty} (-1)^k u[k] \cdot e^{-(n-k)}u[n-k] \end{aligned}$$

For $n < 0$ this evaluates to 0. For $n \geq 0$ this evaluates to

$$\begin{aligned} &= \sum_{k=0}^{n} (-1)^k \cdot e^{-(n-k)} \\ &= e^{(-n)} \sum_{k=0}^{n} (-1)^k \cdot e^k \\ &= e^{(-n)} \sum_{k=0}^{n} (-e)^k \\ &= e^{(-n)} \frac{1 - (-e)^{(n+1)}}{1+e} \\ &= \frac{e^{(-n)} - (-1)^{(n+1)}e}{1+e} \end{aligned}$$

$$\boxed{x[n] * h[n] = \frac{e^{(-n)} - (-1)^{(n+1)}e}{1+e}u[n]}$$

e) $x[n] = (0.5)^n u[n], h[n] = 2^{(-n)}u[-n]$

First note that $2^{(-n)} = 0.5^n$

$$
\begin{aligned}
x[n] * h[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\
&= \sum_{k=-\infty}^{\infty} 0.5^k u[k] \cdot 0.5^{n-k} u[k-n] \\
&= \sum_{k=\max\{0,n\}}^{\infty} 0.5^k \cdot 0.5^{n-k} \\
&= \sum_{k=\max\{0,n\}}^{\infty} 0.5^n \\
&= \infty
\end{aligned}
$$

$\boxed{x[n] * h[n] \text{ diverges}}$

Grading: *20 points, 4 points each*

- For each of (a), (b), (c), (d), and (e):
  - +4: correct answer (-1 for each minor math mistake)
  - +0: empty or invalid